

individuals are fairly certain that their shared content will be viewed by like-minded users, and hence become less disciplined in sharing low-reliability, divisive content. The presence of strategic complementarities once again amplifies this effect, as users recognize that others in their community will tend to share divisive content, and this makes them even more willing to share.

It is notable that Proposition 3, like Theorem 2, implies that greater divisiveness and polarization increase the virality of especially low-reliability content, which is most likely to contain misinformation. These two results together thus imply that echo chambers, greater political polarization and divisive content all exacerbate the circulation of misinformation on social media.

5 Platform Design and Filter Bubbles

We now turn to our second main result: how platform behavior affects misinformation. Consider a collection of social media users with beliefs distributed according to a distribution H . The platform can identify communities of users according to prior ideological beliefs, for example, based on content previously shared or affiliations with ideological groups. In particular, each user is binned into one of k communities, with each community ℓ having a belief distribution H_ℓ with support over $[b_\ell, b_{\ell+1}]$, and where $1 \geq b^{(1)} > b^{(2)} > \dots > b^{(k)} > b^{(k+1)} \geq 0$ (with at least one left-wing and one right-wing community). The size of these bins depends on the platform’s microtargeting technology at identifying users’ ideological beliefs (see, for example, Papakyriakopoulos et al. (2018)). Formally, we let $\varepsilon \equiv \max_\ell (b_{\ell+1} - b_\ell)$, with the interpretation that lower values of ε correspond to better platform technology for identifying ideology.

The platform’s objective is to maximize user engagement, which is equivalent to maximizing content virality (see the definition of content virality in the previous section).¹⁶ The platform does not directly care about whether the content users are engaging with is truthful or contains misinformation.

The platform chooses how content is shared across users. That is, for each article, the platform not only picks the seed agent at $t = 1$ to whom it recommends this article, but also chooses the sharing network—the matrix of link probabilities $\mathbf{P}$.¹⁷ The platform’s choice of $\mathbf{P}$ can be interpreted as its “algorithm” to determine how users are exposed to content circulating in the social media site.

5.1 Optimality of Island Networks and Filter Bubbles

We remind the reader that the island networks of Section 4 are parameterized by three components: the within-island link probability (p_s), the across-island link probability (p_d), and the number of islands (k). As special cases, we have (i) an island model that has *maximal homophily*, where $p_s > 0$ but

¹⁶This objective is rooted in the fact that social media sites, like Facebook, primarily rely on advertising revenue, which becomes more valuable as users increase their activity on the site. For example, 85% of Facebook’s total revenue in 2011 was from advertising, and from 2017-2019, around 98% was (see Andrews (2012) and <https://www.nasdaq.com/articles/what-facebooks-revenue-breakdown-2019-03-28-0>).

Strictly speaking, we are modeling social media platform objectives before the more recent public backlash over misinformation. If the platform faces potential penalties from public backlash or regulators for spreading misinformation, its objective function will change, as we explore in greater detail in Section 6.

¹⁷Facebook’s algorithms may induce different sharing networks depending on features of the article, such as whether it contains cat videos, wedding photos, or political content. See <https://about.fb.com/news/2021/01/how-does-news-feed-predict-what-you-want-to-see/>.